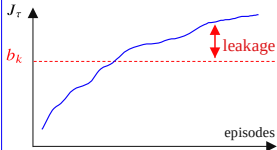


Penalty view

$$\max_{\tau} J_{\tau} - \lambda^{\top} J_c$$

costs $\rightarrow$ objective
 $\tau \rightarrow \infty$

$$J_{\tau} = [J_{c1}, \dots, J_{cN}]^{\top}, \quad \lambda = [\lambda_1, \dots, \lambda_N]^{\top}$$



fixed- λ penalty: some safety tasks reward b_k

weakness: fixed-price leakage

SafeOT

one flow $\cdot$ all budgets

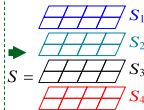
1. Constraints

 C_1 S_1	$C_1 \cdot b_1$
 C_2 S_2	$C_2 \cdot b_2$
 C_3 S_3	$C_3 \cdot b_3$
 H S_4	$H: F_H = 0$

2. One flow

$$\text{one flow}$$

$$S = \min_i S_i$$



$$F\star \rightarrow \lambda\star$$

3. Actor (entropic)

$$\lambda\star + \beta$$

$$\rightarrow \text{actor}$$

$$P =$$

λ_1	λ_2	λ_3
b_1	b_2	b_3
$F\star$	β	λ

safety prices

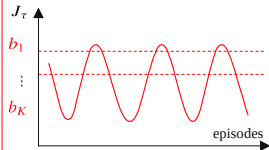
$$P^{\top} \leq S$$

Constraint

$$\max J_{\tau} \quad \text{s.t.} \quad J_c \leq b$$

$$\tau \rightarrow 0$$

$$J_{\tau} = [J_{c1}, \dots, J_{cN}]^{\top}, \quad b = [b_1, \dots, b_N]^{\top}$$



dual updates: safety costs oscillate around b_k

weakness: late / oscillation

$\tau = \infty$

fixed

Lagrangian

one price rule

SafeOT

LP $\varepsilon \rightarrow 0$ / active

$\tau = 0$

penalty

constraints